

IT3212: Data Driven Software

Introduction to the course

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About me



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About you

Kahoot Quiz



About the course -- basics

- 13 lectures
 - 2 hours physical lectures (S2)
- 4 group assignments
 - All are included in the final grade
 - BUT → you have to pass all of them to be able to pass the course

About the course -- lectures

- Pipeline
 - What are they?
 - Examples from SE, IS, HCI
 - How they power software and business development
- Data preprocessing and calibration
 - Importance
 - Techniques
- Knowledge extraction techniques from data

About the course -- lectures

- Basic modelling
 - Statistical processes
- Advanced modelling
 - Combining the processes
- Empowering systems using supervised methods
 - Support vector machines, Random Forest, Gaussian process modelling
 - Temporal modelling

About the course -- lectures

- Empowering systems using reinforcement methods
 - Using no training data
- Empowering systems using LLMs
 - Creating support apps

About the course – group assignments

- Four assignments
 - 18th September → Data Preprocessing
 - 9th October → Image Processing (mostly knowledge extraction)
 - 30th October → Basic and Advanced Modelling
 - 30th November → Reinforcement learning and LLMs
- Passing Grade for all the assignments is 65% of the individual points per assignment

About the course – group assignments

- Four assignments
 - 18th September → Data Preprocessing
 - 9th October → Image Processing (mostly knowledge extraction)
 - 30th October → Basic and Advanced Modelling
 - 30th November → Reinforcement learning and LLMs
- Grading
 - Assignment 1: 20% of the marks in the assignment if passed, 0% if failed
 - Assignment 2: 20% of the marks in the assignment if passed, 0% if failed
 - Assignment 3: 30% of the marks in the assignment if passed, 0% if failed
 - Assignment 4: 30% of the marks in the assignment if passed, 0% if failed

About the course – group assignments

- Resubmission of assignments
 - 30th November → you can resubmit assignments 1, 2, and 3 based on the feedback
 - You will have to include a change log in the form of how have you responded to the feedback on those assignments.
- New Grading for the resubmitted assignments
 - 1.If the score after resubmission improves, then the new score for that assignment is the average of the two grades
 - 2.If the score after resubmission decreases, then the new score for that assignment is the same as the old grade

About the course -- groups

- Make groups of 4 people
 - No less than 3 (3 is exceptional)
 - No more than 5 (5 is exceptional)
- In the assignment and report including the following two things are mandatory
 - Individual contribution
 - Personal reflection
- Deadline for making groups
 - 4th September
 - After this deadline → random allocation
 - There is a forum on canvas to support group formation → **Use It!!**

Can I use online help?

- Yes, but!!!

Summary of Deadlines

- 1st September – enroll in/out of the course
 - ➔ Please contact me if you are having any troubles, last year there were some issues
- 4th September – make groups
- 18th September – Assignment 1
- 9th October – Assignment 2
- 30th October – Assignment 3
- 30th November – Assignment 4

Thank you !!
Questions ??